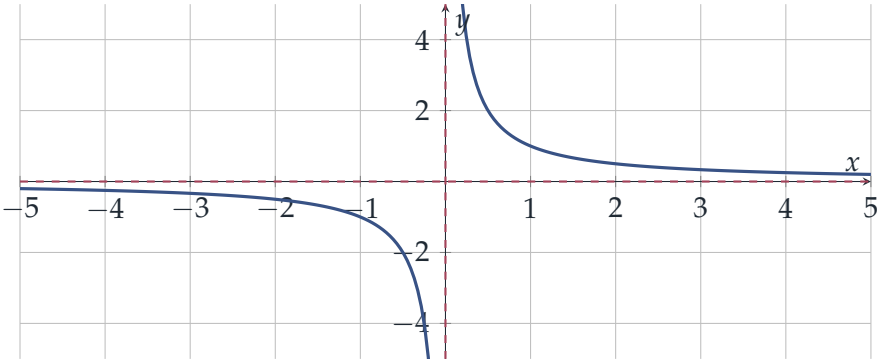
Chapter 4

Reciprocal function

Skills: determine a domain, differentiate, study a sign, draw up a table of variations, represent

4.1 Reference notes

The reciprocal function is defined by $f(x) = \frac{1}{x}$ on $\mathbb{R} \setminus \{0\}$.



Key ideas

$f'(x) = -\frac{1}{x^2} < 0$ for every $x \neq 0$. The reciprocal function is strictly decreasing on each of the intervals $]-\infty, 0[$ and $]0, +\infty[$. The lines $x = 0$ and $y = 0$ are asymptotes.

Common mistake

We cannot write that the reciprocal function is decreasing "on $\mathbb{R} \setminus \{0\}$" without specifying the two intervals. For example $-1 < 1$ but $f(-1) = -1 < 1 = f(1)$, which would contradict the global definition of decreasing.

For a function $g(x) = P(x) + \frac{k}{x}$, proceed in this order: domain; derivative; common denominator and factorisation; sign of the derivative; variations; limits or behaviour at the bounds; useful points; representation.

Worked example

On $]0, +\infty[$, $g(x) = x + \frac{4}{x}$ satisfies $g'(x) = 1 - \frac{4}{x^2} = \frac{(x-2)(x+2)}{x^2}$. The denominator is positive; g' is negative on $]0, 2[$ then positive on $]2, +\infty[$. The function has a minimum $g(2) = 4$.

Session 23 — Definition and graph of the reciprocal function

Reciprocal function

15/12/2026 • B : 2 h – C : 1 h

Objectives of the session

· Determine the domain.

· Draw the curve.

· Use symmetries.

Fluency · 6 to 8 minutes, without calculator

1. Simplify 3 $x + 5 - x + 2$. .....

2. Give the forbidden value of $\dfrac{x+1}{x}$. .....

3. Calculate $\dfrac{1}{5}$ and $-\dfrac{1}{5}$. .....

4. Differentiate 3 $x^2 - 5x + 1$. .....

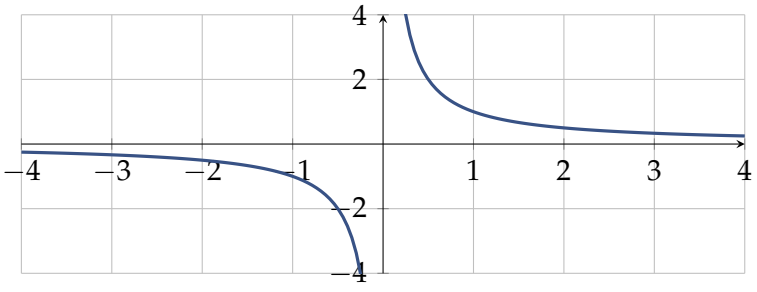
5. Solve $(x-2)(x+1) = 0$. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: determine the domain and draw the graph.

Investigation

Calculate $\dfrac{1}{x}$ for $x = -4, -2, -1, -\dfrac{1}{2}, \dfrac{1}{2}$, 1, 2, 4. Plot the points and describe the symmetries. Why $x = 0$ missing?



Written summary

The reciprocal function is $f(x) = \dfrac{1}{x}$ on $\mathbb{R} \setminus \{0\}$. Its graph is a hyperbola symmetric about at the origin.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Give the domain and the asymptotes of $f(x) = 5 + \frac{7}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study on $]0, +\infty[$ function $g(x) = x + \frac{25}{x}$.

References in the textbook

Extensions: Calao, courses p. 88·93; exercises p. 94, n° 1-5.
session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 101, Exercise 42

The aim of this exercise is to draw the graphs of the functions f and g defined on $\mathbb{R}^*$ by: $f(x) = \frac{2}{x}$ and $g(x) = 4 - \frac{3}{x}$.

1. Complete the table of values for $x \in \{-5; -4; -2; -0{,}5; -0{,}2; -0{,}1; 0{,}1; 0{,}2; 0{,}5; 1; 2; 4; 5\}$.
2. Using a table of values, draw the graphs of the functions f and g.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 235 and the rate or common ratio by 7. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 24 — Behaviour at the endpoints and asymptotes

Reciprocal function

17/12/2026 • B : 1 h – C : 2 h

Objectives of the session

· Describe the behaviour at the endpoints.

· Identify the asymptotes.

· Read a representation.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 5 per cent of 120 .....

2. Calculate $\frac{1}{6}$ and $-\frac{1}{6}$. .....

3. Differentiate 3 $x^2 - 5x + 1$. .....

4. Solve $(x-2)(x+1) = 0$. .....

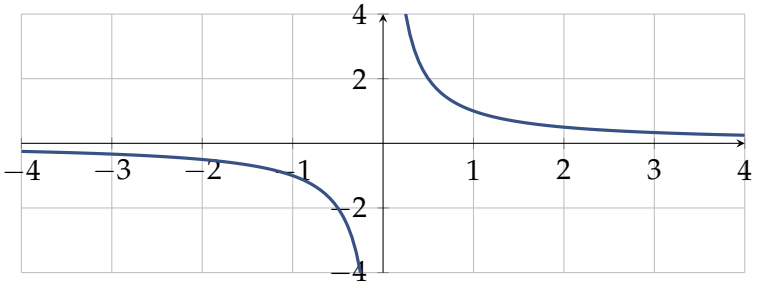
5. Give the forbidden value of $\frac{x+1}{x}$. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: describe the behaviour at the endpoints and identify the asymptotes.

Investigation

Using a table of values close to 0 and for large values of $|x|$, describe $\frac{1}{x}$. Express each behaviour with an asymptote and a sentence.



Written summary

For $f(x) = \frac{1}{x}$, $x = 0$ is a vertical asymptote and $y = 0$ is a horizontal asymptote. The branches approach it approach them without crossing them.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears

in the teacher's edition.

Practice · essentials

1. Give the domain and the asymptotes of $f(x) = 6 + \dfrac{8}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study on $]0, +\infty[$ function $g(x) = x + \dfrac{36}{x}$.

References in the textbook

Extensions: Calao, courses p. 88·93; exercises p. 94, n° 4 to 9.
session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 95, Exercise 16

We want to plot the graph of the function g defined on $\mathbb{R}^*$ by: $g(x) = 3 - \dfrac{2}{x}$.

1. Complete a table of values for

$$x \in \{-1\,000; -500; -200; -100; -50; 50; 100; 200; 500; 1\,000\}.$$

What can we deduce about the behaviour of g in $-\infty$ and at $+\infty$?

2. Complete a table of values for $x = -0{,}5; -0{,}1; -0{,}01; 0{,}01; 0{,}1; 0{,}5$. What can we deduce about the behaviour of g at 0?
3. Copy and complete the table of values from the textbook, then plot the curve of function g.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 240 and the rate or common ratio by 2. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 25 — Derivative and variation

Reciprocal function

05/01/2027 • B : 2 h – C : 1 h

Objectives of the session

· Differentiate the reciprocal function.

· Justify the variations.

· Draw up a table.

Fluency · 6 to 8 minutes, without calculator

1. Write the multiplier for a 10% increase

2. Differentiate 3 $x^2 - 5x + 1$. .....

3. Solve $(x-2)(x+1) = 0$. .....

4. Give the forbidden value of $\dfrac{x+1}{x}$. .....

5. Calculate $\dfrac{1}{7}$ and ~~$-\dfrac{1}{7}$~~. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: differentiate the reciprocal function and justify the variations.

Investigation

From the rate of change of $f(x) = \dfrac{1}{x}$, recover $f'(x) = -\dfrac{1}{x^2}$. Study its sign and check consistency with the graph.

Written summary

For $x \neq 0$, $f'(x) = -\dfrac{1}{x^2} < 0$. The reciprocal function decreases on each of the intervals $]-\infty, 0[$ and $]0, +\infty[$.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Give the domain and the asymptotes of $f(x) = 2 + \dfrac{4}{x}$.

2. Calculate $f'(x)$ and study its sign.

3. Study on $]0, +\infty[$ function $g(x) = x + \dfrac{4}{x}$.

References in the textbook

Extensions: Calao, courses p. 89·93; exercises p. 94, n° 1 to 3, and p. 96, n° 10 to 13. reproduced in this session is identified below by its page and number; the reference allows also to find the figures and digital resources of the textbook.

Calao textbook · p. 94, Exercise 3

Calculate the derivatives of the following functions defined on $\mathbb{R}^*$.

a. $f(x) = 2x - 3 + \dfrac{4}{x}$

b. $g(x) = 3x - 5 - \dfrac{1}{x}$

c. $f(x) = 7x^3 - 2x + \dfrac{1}{x}$

d. $g(x) = 3x^2 - 5x - 2 + \dfrac{4}{x}$

Independent work

Redo practice exercise 2 with the following data: replace the initial value with
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 26 — Combinations with polynomial functions

Reciprocal function

07/01/2027 • B : 1 h – C : 2 h

Objectives of the session

· Differentiate a combination.

· Simplify an expression.

· Organise a study.

Fluency · 6 to 8 minutes, without calculator

1. Simplify 3 $x+8-x+2$. .....

2. Solve $(x-2)(x+1)=0$. .....

3. Give the forbidden value of $\dfrac{x+1}{x}$. .....

4. Calculate $\dfrac{1}{8}$ and $-\dfrac{1}{8}$. .....

5. Differentiate 3 x^2-5x+1. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: differentiate a combination and simplify an expression.

Investigation

A company produces $x>0$ hundred objects and its average cost is $C(x)=2x+30+\dfrac{200}{x}$.
Calculate $C'(x)$, factorise its numerator and find the production level that minimises the cost mean.

Written summary

To study $P(x)+\dfrac{k}{x}$, determine the domain, calculate the derivative, reduce to the same denominator, study its sign then draw up the variations.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Give the domain and the asymptotes of $f(x)=3+\dfrac{5}{x}$.

2. Calculate $f'(x)$ and study its sign.

3. Study on $]0,+\infty[$ function $g(x)=x+\dfrac{9}{x}$.

References in the textbook

Extensions: Calao, p. 96, No. 14-21. The exercise reproduced in this session is identified below. by its page and number; the reference also lets you find the figures and the digital resources of the textbook.

Calao textbook · p. 96, Exercise 20

Let f the function defined on $\mathbb{R}^*$ by $f(x) = 4x + 1 + \frac{9}{x}$.

1. Show that for every real x non-zero: $f'(x) = \frac{(2x-3)(2x+3)}{x^2}$.
2. For every real x non-zero, study the sign of $f'(x)$.
3. Deduce the variation of the function f.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 27 — Sign and factorisation

Reciprocal function

12/01/2027 • B : 2 h – C : 1 h

Objectives of the session

· Factorise.

· Build a sign table.

· Respect the forbidden values.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 20 per cent of 180 . . . . . .

2. Give the forbidden value of $\dfrac{x+1}{x}$. . . . . .

3. Calculate $\dfrac{1}{9}$ and $-\dfrac{1}{9}$. . . . . .

4. Differentiate 3 $x^2 - 5x + 1$. . . . . .

5. Solve $(x-2)(x+1) = 0$. . . . . .

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: factorise and build a sign table.

Investigation

Study the sign of $\dfrac{(x-2)(x+3)}{x}$. Construct the table with the three critical values, distinguish zero and forbidden value, then solve the inequality $\dfrac{(x-2)(x+3)}{x} \geq 0$.

Written summary

In a quotient, the sign depends on the numerator and the denominator. The forbidden values are excluded from the table, even if a factor formally cancels.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Give the domain and the asymptotes of $f(x) = 4 + \dfrac{6}{x}$.

2. Calculate $f'(x)$ and study its sign.

3. Study on $]0, +\infty[$ function $g(x) = x + \dfrac{16}{x}$.

References in the textbook

Extensions: Calao, pp. 96-101. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the resources digital resources of the textbook.

Calao textbook · p. 96, Exercise 21

Let g the function defined on $\mathbb{R}^*$ by $g(x) = x + 10 - \frac{1}{x}$.

1. Show that for every real x non-zero: $g'(x) = \frac{x^2+1}{x^2}$.
2. For every real x non-zero, study the sign of $g'(x)$.
3. Deduce the variation of the function g.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 255 and the rate or common ratio by 5. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 28 — Derivatives of polynomials of degree at most 3

Reciprocal function

14/01/2027 • B : 1 h – C : 2 h

Objectives of the session

· Differentiate a polynomial.

· Study a derivative.

· Deduce variations.

Fluency · 6 to 8 minutes, without calculator

1. Write the multiplier for a 25% increase

2. Calculate $\frac{1}{3}$ and $-\frac{1}{3}$. .....

3. Differentiate 3 $x^2 - 5x + 1$. .....

4. Solve $(x-2)(x+1) = 0$. .....

5. Give the forbidden value of $\frac{x+1}{x}$. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: differentiate a polynomial and study a derivative.

Investigation

For $f(x) = 2x^3 - 3x^2 - 12x + 5$, calculate $f'(x)$, factorise, study its sign and deduce the changes in f.

Written summary

$(ax^3 + bx^2 + cx + d)' = 3ax^2 + 2bx + c$. The sign of the derivative gives the variation of the function.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Give the domain and the asymptotes of $f(x) = 5 + \frac{7}{x}$.

2. Calculate $f'(x)$ and study its sign.

3. Study on $]0, +\infty[$ function $g(x) = x + \frac{25}{x}$.

References in the textbook

Extensions: Calao, pp. 94-96. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the resources digital resources of the textbook.

Calao textbook · p. 94, Exercise 13

Let f the function defined on $\mathbb{R}^*$ by: $f(x) = \dfrac{1}{x}$.

1. Calculate $f'(x)$.
2. Show that for every real x non-zero: $f'(x) = \dfrac{-1}{x^2}$.
3. After studying the sign of $f'(x)$, draw the variation table of f.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 260 and the rate or common ratio by 6. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 29 — Complete study and graph

Reciprocal function

19/01/2027 • B : 2 h – C : 1 h

Objectives of the session

· Carry out a complete study.

· Draw a coherent curve.

· Interpret an extremum.

Fluency · 6 to 8 minutes, without calculator

1. Simplify 3 $x + 4 - x + 2$. .....

2. Differentiate 3 $x^2 - 5x + 1$. .....

3. Solve $(x - 2)(x + 1) = 0$. .....

4. Give the forbidden value of $\dfrac{x+1}{x}$. .....

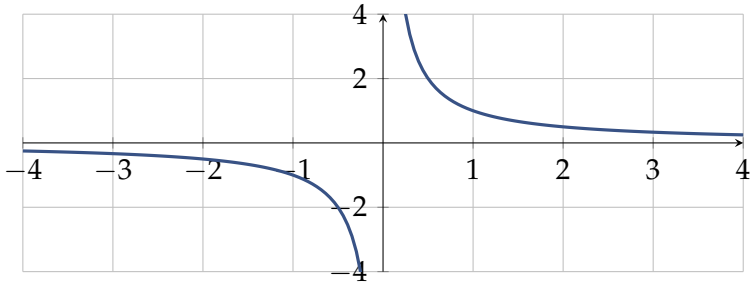
5. Calculate $\dfrac{1}{4}$ and ~~$-\dfrac{1}{4}$~~. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: carry out a complete study and draw a curve coherent.

Investigation

Study fully $g(x) = x + \dfrac{9}{x}$ on $]0, +\infty[$: domain, derivative, sign, variations, minimum, behaviour at the bounds and sketch.



Written summary

A full study includes: domain, derivative, sign, variations, limits or behaviour at the bounds, notable points and a consistent graph.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Give the domain and the asymptotes of $f(x) = 6 + \frac{8}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study on $]0, +\infty[$ function $g(x) = x + \frac{36}{x}$.

References in the textbook

Extensions: Calao, pp. 98-103. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the resources digital resources of the textbook.

Calao textbook · p. 98, Exercise 31

A company specialising in industrial ovens can make between 20 and 80 of them depending on the years. x the number of ovens manufactured annually.
The graph given in the textbook is that of a function C defined on the interval $[20; 80]$. For all integer values of x, $C(x)$ is the unit cost of production mean (in €).

1. What is the average unit production cost when 25 ovens are produced, then when 70 ovens are produced?
2. Which production levels correspond to an average unit cost of 20,000 € ?
3. What is the minimum average unit production cost? What output level does it correspond to?

Independent work

Redo practice exercise 2 with the following data: replace the initial value with
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 30 — Review and assessment

Reciprocal function

21/01/2027 • B : 1 h – C : 2 h

Objectives of the session

· Use domain, derivative and sign.

· Write up a study.

· Correct methodically.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 5 per cent of 100 .....

2. Solve $(x-2)(x+1)=0$. .....

3. Give the forbidden value of $\dfrac{x+1}{x}$. .....

4. Calculate $\dfrac{1}{5}$ and $-\dfrac{1}{5}$. .....

5. Differentiate 3 x^2-5x+1. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: use domain, derivative and sign and write a study.

Investigation

Correct a deliberately flawed study of $h(x)=x+\dfrac{4}{x}$: stated domain $\mathbb{R}$, derivative $1+\dfrac{4}{x^2}$, table with no break at 0. Rewrite a complete solution.

Written summary

Check the four items separately: domain, expression of the derivative, sign of the derivative, variations. An error at one step must not hide the following ones.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Give the domain and the asymptotes of $f(x)=2+\dfrac{4}{x}$.

2. Calculate $f'(x)$ and study its sign.

3. Study on $]0,+\infty[$ function $g(x)=x+\dfrac{4}{x}$.

References in the textbook

Extensions: Calao, pp. 100-103. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the resources digital resources of the textbook.

Calao textbook · p. 98, Exercise 32

Each week, a liquid detergent company estimates that the production cost (in euros) can be modelled by a function C given by: $C(x) = x^2 + 60x + 121$, where x is the volume of detergent (in m ^{3}), $x \in [1;30]$.
Let f the function representing the average cost of production per m ^{3} of liquid detergent.

1. Show that for every $x \in [1;30]$, $f(x) = x + 60 + \dfrac{121}{x}$.
2. Calculate $f'(x)$. Show that for every $x \in [1;30]$, $f'(x) = \dfrac{(x-11)(x+11)}{x^2}$.
3. Draw the variation table of f.
4. What is the minimum average production cost? For what quantity of detergent is it reached ?

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 270 and the rate or common ratio by 2. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.